

1 RQ 1: How much context matters

Table 1: MAE - Train Less Year Linear Pct

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0126	0.0125	0.0125	0.0125	0.0125	0.0124	0.0124	0.0124	0.0124	0.0124
	TimesFM	0.0126	0.0126	0.0126	0.0126	0.0127	0.0126	0.0125	0.0125	0.0125	0.0125
	Naive	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186
	Sundial	0.0127	0.0128	0.0126	0.0125	0.0125	0.0126	0.0124	0.0127	0.0123	0.0125
5d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0125	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0123	0.0124
	TimesFM	0.0127	0.0127	0.0126	0.0128	0.0126	0.0125	0.0125	0.0124	0.0127	0.0126
	Naive	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168
	Sundial	0.0125	0.0127	0.0127	0.0125	0.0127	0.0123	0.0123	0.0128	0.0123	0.0126
21d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0125	0.0124	0.0123	0.0123	0.0123	0.0124	0.0123	0.0123	0.0123	0.0123
	TimesFM	0.0124	0.0128	0.0124	0.0124	0.0124	0.0125	0.0124	0.0124	0.0124	0.0124
	Naive	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177
	Sundial	0.0124	0.0124	0.0123	0.0122	0.0123	0.0126	0.0122	0.0122	0.0123	0.0123
63d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0127	0.0126	0.0124	0.0124	0.0125	0.0125	0.0124	0.0124	0.0124	0.0124
	TimesFM	0.0141	0.0130	0.0124	0.0124	0.0125	0.0125	0.0125	0.0125	0.0125	0.0124
	Naive	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148
	Sundial	0.0124	0.0127	0.0125	0.0126	0.0125	0.0127	0.0122	0.0125	0.0124	0.0121

Table 2: MAPE - Train Less Year Linear Pct

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	134.1783	133.7330	134.0340	125.0435	116.0711	110.8131	109.1055	108.0583	107.7716	107.9814
	TimesFM	138.4949	151.4411	163.0544	152.3967	142.8123	139.0801	129.0691	132.5287	122.1941	119.1947
	Naive	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585
	Sundial	151.2314	153.3525	138.8801	143.2995	146.8679	154.3052	131.2480	156.1884	159.2946	133.2309
5d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	132.5010	132.9009	133.1701	128.1975	120.6333	116.8820	112.6570	111.5997	108.6951	109.0334
	TimesFM	123.4558	124.3091	139.9860	161.3876	132.1685	124.2313	115.9314	108.8188	121.0349	118.8515
	Naive	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349
	Sundial	155.1510	149.8311	136.7785	132.4572	144.1474	131.0509	133.4384	138.9813	131.4930	125.9961
21d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	127.1865	129.4689	126.8714	127.6263	117.9400	113.7937	110.8842	112.1888	109.2723	109.5168
	TimesFM	133.4892	140.6378	129.7492	125.8186	128.3048	134.8368	129.6062	127.9105	128.6644	127.4869
	Naive	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749
	Sundial	138.9118	145.5302	122.7694	124.9110	119.0971	130.8785	118.4452	124.4915	130.4870	113.5047
63d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	147.7205	150.0854	159.6237	147.0657	136.0677	126.1054	115.8900	113.1655	111.7820	112.1789
	TimesFM	318.9884	149.7353	117.4401	120.7300	122.7701	120.7060	118.6446	117.5812	116.5407	116.4882
	Naive	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681
	Sundial	146.6512	163.0543	128.6437	124.2252	134.6259	136.4602	120.4218	132.9171	113.1195	123.0840

Table 3: MAE - Train Less Year Linear

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	19.2725	14.8970	3.7485	3.7485	4.7455	3.7485	3.7485	3.7485	3.7485	3.7485
	Chronos	4.0621	4.1001	4.0399	4.1047	4.0632	4.0491	4.0332	4.0229	4.0715	4.1365
	TimesFM	3.9270	3.9479	3.8567	3.8910	3.8445	3.9019	3.9706	3.9963	3.9568	3.9982
	Naive	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485
	Sundial	4.1523	4.2175	4.2529	4.3367	4.1272	3.9899	3.9870	3.9558	4.0793	4.0549
5d	ARIMA	32.1573	29.3235	6.0328	6.0328	12.3248	6.0328	6.0328	6.0328	6.0328	6.0328
	Chronos	6.4522	6.4674	6.4657	6.5207	6.4679	6.3846	6.3009	6.3582	6.4146	6.4010
	TimesFM	6.2218	6.3779	6.3671	6.2820	6.2056	6.0250	6.2491	6.3240	6.3554	6.1552
	Naive	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328
	Sundial	7.4139	7.3704	7.6708	7.9226	7.6210	6.8434	7.1753	7.0043	7.2897	7.2509
21d	ARIMA	39.4096	40.6093	9.5962	9.5962	30.7089	9.5962	9.5962	9.5962	9.5962	9.5962
	Chronos	10.8410	10.9762	10.5734	10.4479	10.5232	10.2100	10.2530	10.4008	10.5413	10.3870
	TimesFM	12.7218	12.6138	12.2810	12.8690	12.6949	12.3224	11.8834	11.9508	11.5060	11.3798
	Naive	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962
	Sundial	15.4205	15.0609	15.8267	15.8091	15.5818	14.5247	13.9510	14.7045	15.0787	14.9551
63d	ARIMA	43.0837	45.2497	23.2427	23.2427	47.6474	23.2427	23.2427	23.2427	23.2427	23.2427
	Chronos	20.1430	19.9799	20.9526	23.7027	26.1963	24.3371	22.7850	21.8287	21.1070	21.7884
	TimesFM	31.0596	21.5431	26.1750	27.9961	28.1967	26.8554	27.0855	27.3686	27.4123	26.5692
	Naive	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427
	Sundial	33.4506	31.8908	34.6578	35.6201	37.3689	36.8622	32.4022	32.4707	33.3985	32.5286

Table 4: MAPE - Train Less Year Linear

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	5.9137	4.5792	1.2265	1.2265	1.5279	1.2265	1.2265	1.2265	1.2265	1.2265
	Chronos	1.3288	1.3413	1.3238	1.3376	1.3298	1.3258	1.3174	1.3131	1.3303	1.3502
	TimesFM	1.2846	1.2957	1.2607	1.2731	1.2537	1.2772	1.2974	1.3076	1.2923	1.3085
	Naive	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265
	Sundial	1.3623	1.3831	1.3908	1.4161	1.3434	1.2949	1.2942	1.2851	1.3260	1.3166
5d	ARIMA	9.8097	8.9506	1.9748	1.9748	3.8370	1.9748	1.9748	1.9748	1.9748	1.9748
	Chronos	2.1096	2.1147	2.1203	2.1339	2.1169	2.0877	2.0595	2.0775	2.0979	2.0930
	TimesFM	2.0405	2.0998	2.1006	2.0696	2.0353	1.9769	2.0520	2.0761	2.0865	2.0184
	Naive	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748
	Sundial	2.4237	2.3996	2.4982	2.5754	2.4673	2.2103	2.3268	2.2710	2.3634	2.3398
21d	ARIMA	12.0167	12.3618	3.0611	3.0611	9.3681	3.0611	3.0611	3.0611	3.0611	3.0611
	Chronos	3.5284	3.5511	3.4215	3.3462	3.3567	3.2401	3.2520	3.2902	3.3407	3.2919
	TimesFM	4.1248	4.0393	3.9660	4.1425	4.0832	3.9488	3.8283	3.8503	3.6913	3.6600
	Naive	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611
	Sundial	5.0038	4.7735	5.0363	5.0317	4.9761	4.6004	4.4098	4.6720	4.7639	4.7266
63d	ARIMA	13.0838	13.7079	7.1680	7.1680	14.3847	7.1680	7.1680	7.1680	7.1680	7.1680
	Chronos	6.2533	6.1770	6.4723	7.2246	8.0485	7.4067	6.9116	6.5532	6.3408	6.5582
	TimesFM	10.1453	6.5344	7.9999	8.6112	8.6503	8.2120	8.2541	8.3387	8.3517	8.0762
	Naive	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680
	Sundial	10.2129	9.6459	10.4568	10.7176	11.2414	11.0896	9.7751	9.8487	10.0686	9.8594

Table 5: MAE - Train Less Year Log Pct

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0126	0.0126	0.0126	0.0126	0.0125	0.0125	0.0125	0.0124	0.0124	0.0124
	TimesFM	0.0126	0.0126	0.0126	0.0125	0.0125	0.0125	0.0126	0.0126	0.0125	0.0125
	Naive	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186
	Sundial	0.0127	0.0129	0.0130	0.0126	0.0126	0.0125	0.0124	0.0125	0.0124	0.0124
5d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0125	0.0125	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124
	TimesFM	0.0127	0.0128	0.0128	0.0128	0.0126	0.0127	0.0127	0.0125	0.0124	0.0126
	Naive	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168
	Sundial	0.0127	0.0125	0.0127	0.0127	0.0126	0.0128	0.0125	0.0124	0.0124	0.0123
21d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0125	0.0126	0.0125	0.0124	0.0123	0.0123	0.0123	0.0124	0.0123	0.0123
	TimesFM	0.0124	0.0125	0.0125	0.0125	0.0125	0.0124	0.0124	0.0125	0.0124	0.0124
	Naive	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177
	Sundial	0.0125	0.0125	0.0124	0.0124	0.0127	0.0124	0.0121	0.0126	0.0123	0.0126
63d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0127	0.0128	0.0127	0.0126	0.0124	0.0125	0.0125	0.0125	0.0124	0.0124
	TimesFM	0.0141	0.0125	0.0123	0.0125	0.0125	0.0124	0.0125	0.0125	0.0125	0.0125
	Naive	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148
	Sundial	0.0125	0.0124	0.0125	0.0125	0.0129	0.0126	0.0124	0.0124	0.0124	0.0125

Table 6: MAPE - Train Less Year Log Pct

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	134.1783	139.4795	133.1292	140.3792	135.0154	128.9003	119.5021	112.3735	108.2034	107.5991
	TimesFM	138.4949	151.5322	152.6387	159.1098	152.0299	154.9435	144.8455	138.6742	132.7222	119.7643
	Naive	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585
	Sundial	151.1209	158.0745	149.0589	168.5851	135.6755	146.5035	134.8982	142.7565	133.6520	132.9087
5d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	132.5010	135.3439	132.8000	138.5639	132.3414	129.5913	122.2917	117.1648	112.0707	108.3712
	TimesFM	123.4558	128.9045	125.5438	129.9912	143.2647	150.4977	145.7929	123.4827	114.2034	121.6866
	Naive	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349
	Sundial	144.5733	141.4702	169.7803	154.1190	138.0617	147.1717	136.4928	137.2801	157.0378	124.0856
21d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	127.1865	133.2269	131.1772	136.2742	126.3542	128.2717	118.8328	113.4522	112.5307	109.2277
	TimesFM	133.4892	135.2971	135.2444	133.6525	131.5347	129.1051	128.3010	134.2556	126.8169	127.6303
	Naive	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749
	Sundial	128.0995	132.9253	148.3268	138.1501	139.7654	122.7150	118.9470	127.6134	138.7376	129.0908
63d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	147.7205	153.6198	143.5735	161.1749	149.3287	161.1819	144.1267	129.7317	117.5739	112.3116
	TimesFM	318.9884	183.6699	137.4195	135.5582	122.0313	120.1593	120.7581	119.6072	117.1851	116.7288
	Naive	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681
	Sundial	139.1527	163.6215	166.3291	156.6081	141.0663	132.2994	129.5917	125.5668	128.3944	122.2965

Table 7: MAE - Train Less Year Log

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	19.2725	3.7485	17.8878	13.6387	3.7485	3.7485	6.3302	3.7485	3.7485	3.7485
	Chronos	4.0621	4.0563	4.0821	4.0655	4.0377	4.0693	4.1127	4.0681	4.0339	4.1281
	TimesFM	3.9270	3.9644	3.9856	3.9521	3.8420	3.8812	3.8547	3.9145	3.9907	3.9852
	Naive	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485
	Sundial	4.1952	4.1275	4.1544	4.2884	4.1992	4.2776	4.1748	3.9997	3.9938	4.1023
5d	ARIMA	32.1573	6.0328	31.7942	28.0079	6.0328	6.0328	17.0695	6.0328	6.0328	6.0328
	Chronos	6.4522	6.4427	6.4601	6.4586	6.4284	6.4720	6.4834	6.3806	6.3468	6.4114
	TimesFM	6.2218	6.2673	6.2794	6.3797	6.3518	6.3622	6.1755	6.0309	6.3202	6.1657
	Naive	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328
	Sundial	7.3542	7.2596	7.2858	7.4922	7.4297	7.6696	7.7169	7.0477	6.9692	7.2562
21d	ARIMA	39.4096	9.5962	40.6351	40.3912	9.5962	9.5962	37.0610	9.5962	9.5962	9.5962
	Chronos	10.8410	11.1380	11.1287	10.8807	10.5267	10.5695	10.6131	10.2619	10.2681	10.4007
	TimesFM	12.7218	13.1865	12.8878	12.6860	11.9375	12.5745	12.7128	12.2660	11.8870	11.3884
	Naive	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962
	Sundial	15.2122	14.9911	14.6547	14.6855	15.6743	15.9665	15.8535	14.1743	14.2227	15.7278
63d	ARIMA	43.0837	23.2427	45.0593	45.8942	23.2427	23.2427	51.0715	23.2427	23.2427	23.2427
	Chronos	20.1430	21.4644	20.3851	20.3175	21.4915	23.1092	25.2293	24.6453	22.8305	21.7765
	TimesFM	31.0596	31.7390	28.8117	25.1528	26.1020	26.7791	28.0211	26.7634	27.6040	26.6430
	Naive	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427
	Sundial	31.7948	33.9065	30.1360	31.8114	34.5515	34.7829	37.0482	35.2868	32.8635	32.5995

Table 8: MAPE - Train Less Year Log

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	5.9137	1.2265	5.4893	4.1973	1.2265	1.2265	2.0016	1.2265	1.2265	1.2265
	Chronos	1.3288	1.3256	1.3355	1.3311	1.3241	1.3279	1.3405	1.3315	1.3180	1.3476
	TimesFM	1.2846	1.2986	1.3065	1.2985	1.2572	1.2708	1.2560	1.2813	1.3088	1.3043
	Naive	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265
	Sundial	1.3753	1.3548	1.3614	1.4096	1.3738	1.3977	1.3609	1.2989	1.2964	1.3317
5d	ARIMA	9.8097	1.9748	9.6981	8.5527	1.9748	1.9748	5.2590	1.9748	1.9748	1.9748
	Chronos	2.1096	2.1068	2.1130	2.1093	2.1062	2.1157	2.1191	2.0879	2.0762	2.0958
	TimesFM	2.0405	2.0582	2.0637	2.1066	2.0936	2.0918	2.0236	1.9794	2.0821	2.0210
	Naive	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748
	Sundial	2.4000	2.3791	2.3771	2.4433	2.4148	2.4863	2.5074	2.2822	2.2609	2.3478
21d	ARIMA	12.0167	3.0611	12.3759	12.2954	3.0611	3.0611	11.2846	3.0611	3.0611	3.0611
	Chronos	3.5284	3.5969	3.6090	3.5085	3.4065	3.3970	3.3835	3.2579	3.2522	3.2970
	TimesFM	4.1248	4.2649	4.1495	4.0554	3.8511	4.0466	4.0907	3.9303	3.8285	3.6615
	Naive	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611
	Sundial	4.9178	4.8607	4.7058	4.6481	4.9943	5.0701	5.0322	4.4844	4.4967	4.9969
63d	ARIMA	13.0838	7.1680	13.6759	13.9048	7.1680	7.1680	15.4559	7.1680	7.1680	7.1680
	Chronos	6.2533	6.6497	6.3163	6.2782	6.6851	7.0735	7.6533	7.5118	6.8682	6.5650
	TimesFM	10.1453	10.3788	9.2793	7.6543	7.9844	8.1839	8.5927	8.1904	8.4185	8.1016
	Naive	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680
	Sundial	9.7278	10.3260	9.2008	9.6219	10.4327	10.4883	11.1375	10.6235	9.8865	9.8787

Table 9: MAE - Train Restricted Years Pct

Horizon	Method	1y	2y	3y	4y	5y	6y	7y	8y	9y	10y
1d	ARIMA	0.0123	0.0123	0.0123	0.0122	0.0122	0.0122	0.0122	0.0122	0.0122	0.0122
	Chronos	0.0124	0.0124	0.0124	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	TimesFM	0.0125	0.0123	0.0125	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124
	Naive	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186
	Sundial	0.0125	0.0123	0.0121	0.0123	0.0122	0.0122	0.0123	0.0122	0.0123	0.0122
5d	ARIMA	0.0123	0.0123	0.0123	0.0122	0.0122	0.0122	0.0122	0.0123	0.0123	0.0123
	Chronos	0.0124	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	TimesFM	0.0127	0.0126	0.0124	0.0124	0.0124	0.0124	0.0123	0.0123	0.0123	0.0123
	Naive	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168
	Sundial	0.0126	0.0123	0.0122	0.0122	0.0122	0.0123	0.0122	0.0124	0.0125	0.0123
21d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0124	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	TimesFM	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0125	0.0125
	Naive	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177
	Sundial	0.0127	0.0123	0.0121	0.0124	0.0124	0.0124	0.0124	0.0123	0.0124	0.0126
63d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0124	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	TimesFM	0.0125	0.0124	0.0123	0.0124	0.0124	0.0124	0.0123	0.0123	0.0123	0.0123
	Naive	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148
	Sundial	0.0125	0.0125	0.0122	0.0123	0.0122	0.0126	0.0126	0.0124	0.0125	0.0126

Table 10: MAPE - Train Restricted Years Pct

Horizon	Method	1y	2y	3y	4y	5y	6y	7y	8y	9y	10y
1d	ARIMA	100.0000	100.0000	100.0000	117.0533	120.3418	120.2366	118.5998	116.2647	116.3318	116.9548
	Chronos	104.1934	103.5848	103.6555	103.2691	102.9971	101.5214	101.4429	100.8970	100.9207	100.9207
	TimesFM	116.7867	112.8957	114.3041	125.2073	122.4418	124.2768	108.3288	109.2123	109.2409	109.2409
	Naive	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585
	Sundial	122.4922	115.1867	131.8404	128.2986	130.0098	125.2646	124.6425	124.9934	116.6845	113.6216
5d	ARIMA	100.0000	100.0000	100.0000	112.0240	110.8039	111.1422	110.1826	111.2417	110.7549	112.8500
	Chronos	104.9052	108.6503	106.6242	107.9194	108.1942	105.6268	104.6219	103.8299	103.8497	103.8497
	TimesFM	133.9704	129.4419	122.4402	133.3631	135.7388	129.0898	127.8293	124.1571	123.9779	123.9779
	Naive	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349
	Sundial	123.1308	123.3699	131.5948	123.9023	126.5511	123.4469	127.0862	114.9767	126.2855	122.3564
21d	ARIMA	100.0000	100.0000	100.0000	109.9194	108.7572	109.0374	108.2021	108.5775	108.3984	109.4918
	Chronos	108.4636	107.9371	106.7559	108.6453	107.8045	107.0165	106.4038	106.3698	106.3807	106.3807
	TimesFM	122.2589	120.1104	124.1042	122.8145	115.5768	113.6086	121.5208	120.8612	121.1406	121.1406
	Naive	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749
	Sundial	136.3666	112.6410	119.6398	128.5453	127.2375	121.3318	123.9586	124.3354	122.2204	114.0635
63d	ARIMA	100.0000	100.0000	100.0000	110.0507	109.1190	109.6020	108.5764	108.2048	108.2505	108.7890
	Chronos	108.5759	106.8166	106.5247	110.2616	107.1671	106.4238	105.4599	105.5089	105.2983	105.2983
	TimesFM	116.8413	110.2375	108.6242	105.9956	107.6765	109.5049	107.8155	107.2481	107.0414	107.0414
	Naive	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681
	Sundial	135.5120	125.1530	121.0971	117.7993	126.8440	120.9992	119.2775	122.7668	124.6374	115.6296

Table 11: MAE - Train Restricted Years

Horizon	Method	1y	2y	3y	4y	5y	6y	7y	8y	9y	10y
1d	ARIMA	3.7485	3.7485	3.7485	3.7318	3.7316	3.7178	3.7193	3.6933	3.6934	3.6940
	Chronos	3.9963	4.1247	4.3183	4.2923	4.2530	4.4444	4.4569	4.4384	4.4430	4.4430
	TimesFM	3.8884	3.8588	4.1497	4.2709	4.2687	4.2221	4.0621	4.0725	4.0323	4.0323
	Naive	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485
	Sundial	3.9833	3.9415	3.9444	4.0409	4.0772	4.1733	4.1627	4.1612	4.1749	4.2206
5d	ARIMA	6.0328	6.0328	6.0328	6.0590	6.0585	5.9564	5.9691	5.9173	5.9219	5.9258
	Chronos	6.4279	6.6178	6.8515	6.9471	6.9880	7.1729	6.9576	6.9674	6.9671	6.9671
	TimesFM	6.1040	6.1303	6.0291	6.2050	6.2454	6.4963	6.1728	6.5395	6.5644	6.5644
	Naive	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328
	Sundial	6.7512	6.2714	6.6484	6.5860	6.7273	6.6611	6.5580	6.3729	6.5683	6.5006
21d	ARIMA	9.5962	9.5962	9.5962	9.9622	9.9669	9.5705	9.6042	9.5404	9.5567	9.5650
	Chronos	10.6639	10.6342	10.2619	10.3757	10.2844	10.2561	10.1114	10.1109	10.1054	10.1054
	TimesFM	10.1817	9.7962	9.9627	9.6428	9.7391	8.9690	9.8911	9.7028	9.7435	9.7435
	Naive	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962
	Sundial	12.9228	11.8199	11.7792	12.8623	12.0544	11.6833	10.6049	10.9944	11.4374	11.7607
63d	ARIMA	23.2427	23.2427	23.2427	23.5918	23.5859	21.7662	21.9389	21.7914	21.9113	21.9996
	Chronos	18.7126	24.2715	20.1920	22.4571	22.3831	21.9427	22.0661	22.1359	22.0845	22.0845
	TimesFM	24.1446	24.0747	23.5227	23.9451	23.5574	21.6186	22.9539	23.0956	23.0239	23.0239
	Naive	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427
	Sundial	24.9231	25.2164	29.4990	28.9408	30.2930	29.3023	30.3891	28.2717	28.2765	28.5703

Table 12: MAPE - Train Restricted Years

Horizon	Method	1y	2y	3y	4y	5y	6y	7y	8y	9y	10y
1d	ARIMA	1.2265	1.2265	1.2265	1.2218	1.2218	1.2174	1.2179	1.2094	1.2094	1.2096
	Chronos	1.3066	1.3553	1.4162	1.4087	1.3918	1.4565	1.4573	1.4454	1.4478	1.4478
	TimesFM	1.2760	1.2594	1.3468	1.3859	1.3796	1.3716	1.3236	1.3271	1.3100	1.3100
	Naive	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265
	Sundial	1.2896	1.2793	1.2807	1.3204	1.3342	1.3637	1.3575	1.3555	1.3571	1.3721
5d	ARIMA	1.9748	1.9748	1.9748	1.9843	1.9842	1.9521	1.9562	1.9384	1.9398	1.9411
	Chronos	2.1037	2.1727	2.2491	2.2851	2.2953	2.3557	2.2827	2.2811	2.2810	2.2810
	TimesFM	2.0028	1.9972	1.9796	2.0351	2.0400	2.1250	2.0076	2.1145	2.1248	2.1248
	Naive	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748
	Sundial	2.1710	2.0258	2.1635	2.1682	2.2137	2.1879	2.1509	2.0760	2.1485	2.1234
21d	ARIMA	3.0611	3.0611	3.0611	3.1722	3.1736	3.0426	3.0537	3.0296	3.0351	3.0382
	Chronos	3.3894	3.3920	3.2775	3.3475	3.3364	3.3333	3.2751	3.2698	3.2670	3.2670
	TimesFM	3.2359	3.1178	3.1905	3.0837	3.1279	2.8674	3.1594	3.1121	3.1292	3.1292
	Naive	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611
	Sundial	4.0683	3.6599	3.8193	4.2684	3.9659	3.8185	3.4312	3.5583	3.6799	3.8132
63d	ARIMA	7.1680	7.1680	7.1680	7.2710	7.2695	6.6920	6.7471	6.7038	6.7419	6.7700
	Chronos	5.6701	7.7146	6.3448	7.1901	7.1807	7.0379	7.0446	7.0403	7.0199	7.0199
	TimesFM	7.3662	7.3086	7.2656	7.3976	7.3101	6.6426	7.1514	7.2008	7.1748	7.1748
	Naive	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680
	Sundial	7.4388	7.4502	9.4200	9.5580	9.7080	9.3957	9.6126	8.9511	8.9295	8.9716

2 RQ 2: Impact of going to higher frequency

Table 13: MAE - Train Less Year Linear Pct

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0126	0.0125	0.0125	0.0125	0.0125	0.0124	0.0124	0.0124	0.0124	0.0124
	TimesFM	0.0126	0.0126	0.0126	0.0126	0.0127	0.0126	0.0125	0.0125	0.0125	0.0125
	Naive	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186
	Sundial	0.0127	0.0128	0.0126	0.0125	0.0125	0.0126	0.0124	0.0127	0.0123	0.0125
5d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0125	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0123	0.0124
	TimesFM	0.0127	0.0127	0.0126	0.0128	0.0126	0.0125	0.0125	0.0124	0.0127	0.0126
	Naive	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168
	Sundial	0.0125	0.0127	0.0127	0.0125	0.0127	0.0123	0.0123	0.0128	0.0123	0.0126
21d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0125	0.0124	0.0123	0.0123	0.0123	0.0124	0.0123	0.0123	0.0123	0.0123
	TimesFM	0.0124	0.0128	0.0124	0.0124	0.0124	0.0125	0.0124	0.0124	0.0124	0.0124
	Naive	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177
	Sundial	0.0124	0.0124	0.0123	0.0122	0.0123	0.0126	0.0122	0.0122	0.0123	0.0123
63d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0127	0.0126	0.0124	0.0124	0.0125	0.0125	0.0124	0.0124	0.0124	0.0124
	TimesFM	0.0141	0.0130	0.0124	0.0124	0.0125	0.0125	0.0125	0.0125	0.0125	0.0124
	Naive	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148
	Sundial	0.0124	0.0127	0.0125	0.0126	0.0125	0.0127	0.0122	0.0125	0.0124	0.0121

Table 14: MAPE - Train Less Year Linear Pct

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	134.1783	133.7330	134.0340	125.0435	116.0711	110.8131	109.1055	108.0583	107.7716	107.9814
	TimesFM	138.4949	151.4411	163.0544	152.3967	142.8123	139.0801	129.0691	132.5287	122.1941	119.1947
	Naive	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585
	Sundial	151.2314	153.3525	138.8801	143.2995	146.8679	154.3052	131.2480	156.1884	159.2946	133.2309
5d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	132.5010	132.9009	133.1701	128.1975	120.6333	116.8820	112.6570	111.5997	108.6951	109.0334
	TimesFM	123.4558	124.3091	139.9860	161.3876	132.1685	124.2313	115.9314	108.8188	121.0349	118.8515
	Naive	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349
	Sundial	155.1510	149.8311	136.7785	132.4572	144.1474	131.0509	133.4384	138.9813	131.4930	125.9961
21d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	127.1865	129.4689	126.8714	127.6263	117.9400	113.7937	110.8842	112.1888	109.2723	109.5168
	TimesFM	133.4892	140.6378	129.7492	125.8186	128.3048	134.8368	129.6062	127.9105	128.6644	127.4869
	Naive	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749
	Sundial	138.9118	145.5302	122.7694	124.9110	119.0971	130.8785	118.4452	124.4915	130.4870	113.5047
63d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	147.7205	150.0854	159.6237	147.0657	136.0677	126.1054	115.8900	113.1655	111.7820	112.1789
	TimesFM	318.9884	149.7353	117.4401	120.7300	122.7701	120.7060	118.6446	117.5812	116.5407	116.4882
	Naive	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681
	Sundial	146.6512	163.0543	128.6437	124.2252	134.6259	136.4602	120.4218	132.9171	113.1195	123.0840

Table 15: MAE - Train Less Year Linear

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	19.2725	14.8970	3.7485	3.7485	4.7455	3.7485	3.7485	3.7485	3.7485	3.7485
	Chronos	4.0621	4.1001	4.0399	4.1047	4.0632	4.0491	4.0332	4.0229	4.0715	4.1365
	TimesFM	3.9270	3.9479	3.8567	3.8910	3.8445	3.9019	3.9706	3.9963	3.9568	3.9982
	Naive	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485
	Sundial	4.1523	4.2175	4.2529	4.3367	4.1272	3.9899	3.9870	3.9558	4.0793	4.0549
5d	ARIMA	32.1573	29.3235	6.0328	6.0328	12.3248	6.0328	6.0328	6.0328	6.0328	6.0328
	Chronos	6.4522	6.4674	6.4657	6.5207	6.4679	6.3846	6.3009	6.3582	6.4146	6.4010
	TimesFM	6.2218	6.3779	6.3671	6.2820	6.2056	6.0250	6.2491	6.3240	6.3554	6.1552
	Naive	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328
	Sundial	7.4139	7.3704	7.6708	7.9226	7.6210	6.8434	7.1753	7.0043	7.2897	7.2509
21d	ARIMA	39.4096	40.6093	9.5962	9.5962	30.7089	9.5962	9.5962	9.5962	9.5962	9.5962
	Chronos	10.8410	10.9762	10.5734	10.4479	10.5232	10.2100	10.2530	10.4008	10.5413	10.3870
	TimesFM	12.7218	12.6138	12.2810	12.8690	12.6949	12.3224	11.8834	11.9508	11.5060	11.3798
	Naive	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962
	Sundial	15.4205	15.0609	15.8267	15.8091	15.5818	14.5247	13.9510	14.7045	15.0787	14.9551
63d	ARIMA	43.0837	45.2497	23.2427	23.2427	47.6474	23.2427	23.2427	23.2427	23.2427	23.2427
	Chronos	20.1430	19.9799	20.9526	23.7027	26.1963	24.3371	22.7850	21.8287	21.1070	21.7884
	TimesFM	31.0596	21.5431	26.1750	27.9961	28.1967	26.8554	27.0855	27.3686	27.4123	26.5692
	Naive	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427
	Sundial	33.4506	31.8908	34.6578	35.6201	37.3689	36.8622	32.4022	32.4707	33.3985	32.5286

Table 16: MAPE - Train Less Year Linear

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	5.9137	4.5792	1.2265	1.2265	1.5279	1.2265	1.2265	1.2265	1.2265	1.2265
	Chronos	1.3288	1.3413	1.3238	1.3376	1.3298	1.3258	1.3174	1.3131	1.3303	1.3502
	TimesFM	1.2846	1.2957	1.2607	1.2731	1.2537	1.2772	1.2974	1.3076	1.2923	1.3085
	Naive	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265
	Sundial	1.3623	1.3831	1.3908	1.4161	1.3434	1.2949	1.2942	1.2851	1.3260	1.3166
5d	ARIMA	9.8097	8.9506	1.9748	1.9748	3.8370	1.9748	1.9748	1.9748	1.9748	1.9748
	Chronos	2.1096	2.1147	2.1203	2.1339	2.1169	2.0877	2.0595	2.0775	2.0979	2.0930
	TimesFM	2.0405	2.0998	2.1006	2.0696	2.0353	1.9769	2.0520	2.0761	2.0865	2.0184
	Naive	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748
	Sundial	2.4237	2.3996	2.4982	2.5754	2.4673	2.2103	2.3268	2.2710	2.3634	2.3398
21d	ARIMA	12.0167	12.3618	3.0611	3.0611	9.3681	3.0611	3.0611	3.0611	3.0611	3.0611
	Chronos	3.5284	3.5511	3.4215	3.3462	3.3567	3.2401	3.2520	3.2902	3.3407	3.2919
	TimesFM	4.1248	4.0393	3.9660	4.1425	4.0832	3.9488	3.8283	3.8503	3.6913	3.6600
	Naive	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611
	Sundial	5.0038	4.7735	5.0363	5.0317	4.9761	4.6004	4.4098	4.6720	4.7639	4.7266
63d	ARIMA	13.0838	13.7079	7.1680	7.1680	14.3847	7.1680	7.1680	7.1680	7.1680	7.1680
	Chronos	6.2533	6.1770	6.4723	7.2246	8.0485	7.4067	6.9116	6.5532	6.3408	6.5582
	TimesFM	10.1453	6.5344	7.9999	8.6112	8.6503	8.2120	8.2541	8.3387	8.3517	8.0762
	Naive	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680
	Sundial	10.2129	9.6459	10.4568	10.7176	11.2414	11.0896	9.7751	9.8487	10.0686	9.8594

Table 17: MAE - Train Less Year Log Pct

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0126	0.0126	0.0126	0.0126	0.0125	0.0125	0.0125	0.0124	0.0124	0.0124
	TimesFM	0.0126	0.0126	0.0126	0.0125	0.0125	0.0125	0.0126	0.0126	0.0125	0.0125
	Naive	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186
	Sundial	0.0127	0.0129	0.0130	0.0126	0.0126	0.0125	0.0124	0.0125	0.0124	0.0124
5d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0125	0.0125	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124
	TimesFM	0.0127	0.0128	0.0128	0.0128	0.0126	0.0127	0.0127	0.0125	0.0124	0.0126
	Naive	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168
	Sundial	0.0127	0.0125	0.0127	0.0127	0.0126	0.0128	0.0125	0.0124	0.0124	0.0123
21d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0125	0.0126	0.0125	0.0124	0.0123	0.0123	0.0123	0.0124	0.0123	0.0123
	TimesFM	0.0124	0.0125	0.0125	0.0125	0.0125	0.0124	0.0124	0.0125	0.0124	0.0124
	Naive	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177
	Sundial	0.0125	0.0125	0.0124	0.0124	0.0127	0.0124	0.0121	0.0126	0.0123	0.0126
63d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0127	0.0128	0.0127	0.0126	0.0124	0.0125	0.0125	0.0125	0.0124	0.0124
	TimesFM	0.0141	0.0125	0.0123	0.0125	0.0125	0.0124	0.0125	0.0125	0.0125	0.0125
	Naive	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148
	Sundial	0.0125	0.0124	0.0125	0.0125	0.0129	0.0126	0.0124	0.0124	0.0124	0.0125

Table 18: MAPE - Train Less Year Log Pct

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	134.1783	139.4795	133.1292	140.3792	135.0154	128.9003	119.5021	112.3735	108.2034	107.5991
	TimesFM	138.4949	151.5322	152.6387	159.1098	152.0299	154.9435	144.8455	138.6742	132.7222	119.7643
	Naive	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585
	Sundial	151.1209	158.0745	149.0589	168.5851	135.6755	146.5035	134.8982	142.7565	133.6520	132.9087
5d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	132.5010	135.3439	132.8000	138.5639	132.3414	129.5913	122.2917	117.1648	112.0707	108.3712
	TimesFM	123.4558	128.9045	125.5438	129.9912	143.2647	150.4977	145.7929	123.4827	114.2034	121.6866
	Naive	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349
	Sundial	144.5733	141.4702	169.7803	154.1190	138.0617	147.1717	136.4928	137.2801	157.0378	124.0856
21d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	127.1865	133.2269	131.1772	136.2742	126.3542	128.2717	118.8328	113.4522	112.5307	109.2277
	TimesFM	133.4892	135.2971	135.2444	133.6525	131.5347	129.1051	128.3010	134.2556	126.8169	127.6303
	Naive	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749
	Sundial	128.0995	132.9253	148.3268	138.1501	139.7654	122.7150	118.9470	127.6134	138.7376	129.0908
63d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	147.7205	153.6198	143.5735	161.1749	149.3287	161.1819	144.1267	129.7317	117.5739	112.3116
	TimesFM	318.9884	183.6699	137.4195	135.5582	122.0313	120.1593	120.7581	119.6072	117.1851	116.7288
	Naive	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681
	Sundial	139.1527	163.6215	166.3291	156.6081	141.0663	132.2994	129.5917	125.5668	128.3944	122.2965

Table 19: MAE - Train Less Year Log

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	19.2725	3.7485	17.8878	13.6387	3.7485	3.7485	6.3302	3.7485	3.7485	3.7485
	Chronos	4.0621	4.0563	4.0821	4.0655	4.0377	4.0693	4.1127	4.0681	4.0339	4.1281
	TimesFM	3.9270	3.9644	3.9856	3.9521	3.8420	3.8812	3.8547	3.9145	3.9907	3.9852
	Naive	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485
	Sundial	4.1952	4.1275	4.1544	4.2884	4.1992	4.2776	4.1748	3.9997	3.9938	4.1023
5d	ARIMA	32.1573	6.0328	31.7942	28.0079	6.0328	6.0328	17.0695	6.0328	6.0328	6.0328
	Chronos	6.4522	6.4427	6.4601	6.4586	6.4284	6.4720	6.4834	6.3806	6.3468	6.4114
	TimesFM	6.2218	6.2673	6.2794	6.3797	6.3518	6.3622	6.1755	6.0309	6.3202	6.1657
	Naive	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328
	Sundial	7.3542	7.2596	7.2858	7.4922	7.4297	7.6696	7.7169	7.0477	6.9692	7.2562
21d	ARIMA	39.4096	9.5962	40.6351	40.3912	9.5962	9.5962	37.0610	9.5962	9.5962	9.5962
	Chronos	10.8410	11.1380	11.1287	10.8807	10.5267	10.5695	10.6131	10.2619	10.2681	10.4007
	TimesFM	12.7218	13.1865	12.8878	12.6860	11.9375	12.5745	12.7128	12.2660	11.8870	11.3884
	Naive	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962
	Sundial	15.2122	14.9911	14.6547	14.6855	15.6743	15.9665	15.8535	14.1743	14.2227	15.7278
63d	ARIMA	43.0837	23.2427	45.0593	45.8942	23.2427	23.2427	51.0715	23.2427	23.2427	23.2427
	Chronos	20.1430	21.4644	20.3851	20.3175	21.4915	23.1092	25.2293	24.6453	22.8305	21.7765
	TimesFM	31.0596	31.7390	28.8117	25.1528	26.1020	26.7791	28.0211	26.7634	27.6040	26.6430
	Naive	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427
	Sundial	31.7948	33.9065	30.1360	31.8114	34.5515	34.7829	37.0482	35.2868	32.8635	32.5995

Table 20: MAPE - Train Less Year Log

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	5.9137	1.2265	5.4893	4.1973	1.2265	1.2265	2.0016	1.2265	1.2265	1.2265
	Chronos	1.3288	1.3256	1.3355	1.3311	1.3241	1.3279	1.3405	1.3315	1.3180	1.3476
	TimesFM	1.2846	1.2986	1.3065	1.2985	1.2572	1.2708	1.2560	1.2813	1.3088	1.3043
	Naive	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265
	Sundial	1.3753	1.3548	1.3614	1.4096	1.3738	1.3977	1.3609	1.2989	1.2964	1.3317
5d	ARIMA	9.8097	1.9748	9.6981	8.5527	1.9748	1.9748	5.2590	1.9748	1.9748	1.9748
	Chronos	2.1096	2.1068	2.1130	2.1093	2.1062	2.1157	2.1191	2.0879	2.0762	2.0958
	TimesFM	2.0405	2.0582	2.0637	2.1066	2.0936	2.0918	2.0236	1.9794	2.0821	2.0210
	Naive	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748
	Sundial	2.4000	2.3791	2.3771	2.4433	2.4148	2.4863	2.5074	2.2822	2.2609	2.3478
21d	ARIMA	12.0167	3.0611	12.3759	12.2954	3.0611	3.0611	11.2846	3.0611	3.0611	3.0611
	Chronos	3.5284	3.5969	3.6090	3.5085	3.4065	3.3970	3.3835	3.2579	3.2522	3.2970
	TimesFM	4.1248	4.2649	4.1495	4.0554	3.8511	4.0466	4.0907	3.9303	3.8285	3.6615
	Naive	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611
	Sundial	4.9178	4.8607	4.7058	4.6481	4.9943	5.0701	5.0322	4.4844	4.4967	4.9969
63d	ARIMA	13.0838	7.1680	13.6759	13.9048	7.1680	7.1680	15.4559	7.1680	7.1680	7.1680
	Chronos	6.2533	6.6497	6.3163	6.2782	6.6851	7.0735	7.6533	7.5118	6.8682	6.5650
	TimesFM	10.1453	10.3788	9.2793	7.6543	7.9844	8.1839	8.5927	8.1904	8.4185	8.1016
	Naive	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680
	Sundial	9.7278	10.3260	9.2008	9.6219	10.4327	10.4883	11.1375	10.6235	9.8865	9.8787

Table 21: MAE - Train Restricted Years Pct

Horizon	Method	1y	2y	3y	4y	5y	6y	7y	8y	9y	10y
1d	ARIMA	0.0123	0.0123	0.0123	0.0122	0.0122	0.0122	0.0122	0.0122	0.0122	0.0122
	Chronos	0.0124	0.0124	0.0124	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	TimesFM	0.0125	0.0123	0.0125	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124
	Naive	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186
	Sundial	0.0125	0.0123	0.0121	0.0123	0.0122	0.0122	0.0123	0.0122	0.0123	0.0122
5d	ARIMA	0.0123	0.0123	0.0123	0.0122	0.0122	0.0122	0.0122	0.0123	0.0123	0.0123
	Chronos	0.0124	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	TimesFM	0.0127	0.0126	0.0124	0.0124	0.0124	0.0124	0.0123	0.0123	0.0123	0.0123
	Naive	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168	0.0168
	Sundial	0.0126	0.0123	0.0122	0.0122	0.0122	0.0123	0.0122	0.0124	0.0125	0.0123
21d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0124	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	TimesFM	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0124	0.0125	0.0125
	Naive	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177	0.0177
	Sundial	0.0127	0.0123	0.0121	0.0124	0.0124	0.0124	0.0124	0.0123	0.0124	0.0126
63d	ARIMA	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	Chronos	0.0124	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123	0.0123
	TimesFM	0.0125	0.0124	0.0123	0.0124	0.0124	0.0124	0.0123	0.0123	0.0123	0.0123
	Naive	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148
	Sundial	0.0125	0.0125	0.0122	0.0123	0.0122	0.0126	0.0126	0.0124	0.0125	0.0126

Table 22: MAPE - Train Restricted Years Pct

Horizon	Method	1y	2y	3y	4y	5y	6y	7y	8y	9y	10y
1d	ARIMA	100.0000	100.0000	100.0000	117.0533	120.3418	120.2366	118.5998	116.2647	116.3318	116.9548
	Chronos	104.1934	103.5848	103.6555	103.2691	102.9971	101.5214	101.4429	100.8970	100.9207	100.9207
	TimesFM	116.7867	112.8957	114.3041	125.2073	122.4418	124.2768	108.3288	109.2123	109.2409	109.2409
	Naive	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585	343.9585
	Sundial	122.4922	115.1867	131.8404	128.2986	130.0098	125.2646	124.6425	124.9934	116.6845	113.6216
5d	ARIMA	100.0000	100.0000	100.0000	112.0240	110.8039	111.1422	110.1826	111.2417	110.7549	112.8500
	Chronos	104.9052	108.6503	106.6242	107.9194	108.1942	105.6268	104.6219	103.8299	103.8497	103.8497
	TimesFM	133.9704	129.4419	122.4402	133.3631	135.7388	129.0898	127.8293	124.1571	123.9779	123.9779
	Naive	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349	312.7349
	Sundial	123.1308	123.3699	131.5948	123.9023	126.5511	123.4469	127.0862	114.9767	126.2855	122.3564
21d	ARIMA	100.0000	100.0000	100.0000	109.9194	108.7572	109.0374	108.2021	108.5775	108.3984	109.4918
	Chronos	108.4636	107.9371	106.7559	108.6453	107.8045	107.0165	106.4038	106.3698	106.3807	106.3807
	TimesFM	122.2589	120.1104	124.1042	122.8145	115.5768	113.6086	121.5208	120.8612	121.1406	121.1406
	Naive	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749	531.0749
	Sundial	136.3666	112.6410	119.6398	128.5453	127.2375	121.3318	123.9586	124.3354	122.2204	114.0635
63d	ARIMA	100.0000	100.0000	100.0000	110.0507	109.1190	109.6020	108.5764	108.2048	108.2505	108.7890
	Chronos	108.5759	106.8166	106.5247	110.2616	107.1671	106.4238	105.4599	105.5089	105.2983	105.2983
	TimesFM	116.8413	110.2375	108.6242	105.9956	107.6765	109.5049	107.8155	107.2481	107.0414	107.0414
	Naive	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681	336.7681
	Sundial	135.5120	125.1530	121.0971	117.7993	126.8440	120.9992	119.2775	122.7668	124.6374	115.6296

Table 23: MAE - Train Restricted Years

Horizon	Method	1y	2y	3y	4y	5y	6y	7y	8y	9y	10y
1d	ARIMA	3.7485	3.7485	3.7485	3.7318	3.7316	3.7178	3.7193	3.6933	3.6934	3.6940
	Chronos	3.9963	4.1247	4.3183	4.2923	4.2530	4.4444	4.4569	4.4384	4.4430	4.4430
	TimesFM	3.8884	3.8588	4.1497	4.2709	4.2687	4.2221	4.0621	4.0725	4.0323	4.0323
	Naive	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485	3.7485
	Sundial	3.9833	3.9415	3.9444	4.0409	4.0772	4.1733	4.1627	4.1612	4.1749	4.2206
5d	ARIMA	6.0328	6.0328	6.0328	6.0590	6.0585	5.9564	5.9691	5.9173	5.9219	5.9258
	Chronos	6.4279	6.6178	6.8515	6.9471	6.9880	7.1729	6.9576	6.9674	6.9671	6.9671
	TimesFM	6.1040	6.1303	6.0291	6.2050	6.2454	6.4963	6.1728	6.5395	6.5644	6.5644
	Naive	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328	6.0328
	Sundial	6.7512	6.2714	6.6484	6.5860	6.7273	6.6611	6.5580	6.3729	6.5683	6.5006
21d	ARIMA	9.5962	9.5962	9.5962	9.9622	9.9669	9.5705	9.6042	9.5404	9.5567	9.5650
	Chronos	10.6639	10.6342	10.2619	10.3757	10.2844	10.2561	10.1114	10.1109	10.1054	10.1054
	TimesFM	10.1817	9.7962	9.9627	9.6428	9.7391	8.9690	9.8911	9.7028	9.7435	9.7435
	Naive	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962	9.5962
	Sundial	12.9228	11.8199	11.7792	12.8623	12.0544	11.6833	10.6049	10.9944	11.4374	11.7607
63d	ARIMA	23.2427	23.2427	23.2427	23.5918	23.5859	21.7662	21.9389	21.7914	21.9113	21.9996
	Chronos	18.7126	24.2715	20.1920	22.4571	22.3831	21.9427	22.0661	22.1359	22.0845	22.0845
	TimesFM	24.1446	24.0747	23.5227	23.9451	23.5574	21.6186	22.9539	23.0956	23.0239	23.0239
	Naive	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427	23.2427
	Sundial	24.9231	25.2164	29.4990	28.9408	30.2930	29.3023	30.3891	28.2717	28.2765	28.5703

Table 24: MAPE - Train Restricted Years

Horizon	Method	1y	2y	3y	4y	5y	6y	7y	8y	9y	10y
1d	ARIMA	1.2265	1.2265	1.2265	1.2218	1.2218	1.2174	1.2179	1.2094	1.2094	1.2096
	Chronos	1.3066	1.3553	1.4162	1.4087	1.3918	1.4565	1.4573	1.4454	1.4478	1.4478
	TimesFM	1.2760	1.2594	1.3468	1.3859	1.3796	1.3716	1.3236	1.3271	1.3100	1.3100
	Naive	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265	1.2265
	Sundial	1.2896	1.2793	1.2807	1.3204	1.3342	1.3637	1.3575	1.3555	1.3571	1.3721
5d	ARIMA	1.9748	1.9748	1.9748	1.9843	1.9842	1.9521	1.9562	1.9384	1.9398	1.9411
	Chronos	2.1037	2.1727	2.2491	2.2851	2.2953	2.3557	2.2827	2.2811	2.2810	2.2810
	TimesFM	2.0028	1.9972	1.9796	2.0351	2.0400	2.1250	2.0076	2.1145	2.1248	2.1248
	Naive	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748	1.9748
	Sundial	2.1710	2.0258	2.1635	2.1682	2.2137	2.1879	2.1509	2.0760	2.1485	2.1234
21d	ARIMA	3.0611	3.0611	3.0611	3.1722	3.1736	3.0426	3.0537	3.0296	3.0351	3.0382
	Chronos	3.3894	3.3920	3.2775	3.3475	3.3364	3.3333	3.2751	3.2698	3.2670	3.2670
	TimesFM	3.2359	3.1178	3.1905	3.0837	3.1279	2.8674	3.1594	3.1121	3.1292	3.1292
	Naive	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611	3.0611
	Sundial	4.0683	3.6599	3.8193	4.2684	3.9659	3.8185	3.4312	3.5583	3.6799	3.8132
63d	ARIMA	7.1680	7.1680	7.1680	7.2710	7.2695	6.6920	6.7471	6.7038	6.7419	6.7700
	Chronos	5.6701	7.7146	6.3448	7.1901	7.1807	7.0379	7.0446	7.0403	7.0199	7.0199
	TimesFM	7.3662	7.3086	7.2656	7.3976	7.3101	6.6426	7.1514	7.2008	7.1748	7.1748
	Naive	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680	7.1680
	Sundial	7.4388	7.4502	9.4200	9.5580	9.7080	9.3957	9.6126	8.9511	8.9295	8.9716

3 RQ 3: Cases of financial disruption

Table 25: MAE - Rq3 Covid Crash 2020 Linear Pct

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	0.0117	0.0117	0.0117	0.0116	0.0116	0.0116	0.0116	0.0116	0.0116	0.0116
	Chronos	0.0119	0.0120	0.0120	0.0119	0.0119	0.0119	0.0119	0.0119	0.0119	0.0119
	TimesFM	0.0123	0.0123	0.0122	0.0121	0.0121	0.0121	0.0122	0.0123	0.0121	0.0121
	Naive	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186
	Sundial	0.0120	0.0119	0.0117	0.0117	0.0118	0.0120	0.0117	0.0119	0.0118	0.0120

Table 26: MAPE - Rq3 Covid Crash 2020 Linear Pct

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	179.3309	180.7942	149.7141	155.4025	148.3716	148.1740	146.9073	143.6966	140.4699	139.2579
	Chronos	114.0797	106.0699	105.3575	103.7013	102.0420	103.2446	102.8260	103.4092	103.6824	103.5411
	TimesFM	136.8285	140.4527	132.3024	125.3974	135.7149	132.2802	136.6939	142.6412	141.7005	137.0197
	Naive	401.3719	401.3719	401.3719	401.3719	401.3719	401.3719	401.3719	401.3719	401.3719	401.3719
	Sundial	123.2544	133.5280	139.4844	113.1549	125.9786	132.9738	119.8935	143.7151	140.0698	128.1822

Table 27: MAE - Rq3 Covid Crash 2020 Linear

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	1.8219	1.8210	1.8196	1.8094	1.8177	1.8076	1.8065	1.8054	1.8043	1.8039
	Chronos	1.9103	1.9142	1.9507	2.0058	2.0226	1.9778	1.9639	1.9409	1.9187	1.9175
	TimesFM	1.9194	1.8914	1.9046	1.8943	1.9230	1.8695	1.8878	1.9122	1.8901	1.8524
	Naive	1.8353	1.8353	1.8353	1.8353	1.8353	1.8353	1.8353	1.8353	1.8353	1.8353
	Sundial	1.8418	1.8337	1.8112	1.8062	1.8055	1.7946	1.8121	1.7908	1.8121	1.8132

Table 28: MAPE - Rq3 Covid Crash 2020 Linear

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	1.1709	1.1719	1.1711	1.1652	1.1703	1.1640	1.1634	1.1626	1.1619	1.1616
	Chronos	1.2282	1.2294	1.2536	1.2901	1.3015	1.2727	1.2628	1.2475	1.2328	1.2320
	TimesFM	1.2358	1.2194	1.2260	1.2170	1.2389	1.2015	1.2139	1.2259	1.2134	1.1936
	Naive	1.1821	1.1821	1.1821	1.1821	1.1821	1.1821	1.1821	1.1821	1.1821	1.1821
	Sundial	1.1876	1.1843	1.1687	1.1654	1.1636	1.1561	1.1673	1.1538	1.1677	1.1679

Table 29: MAE - Rq3 Covid Crash 2020 Log Pct

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	0.0117	0.0117	0.0118	0.0118	0.0117	0.0116	0.0116	0.0115	0.0116	0.0116
	Chronos	0.0119	0.0119	0.0119	0.0120	0.0120	0.0119	0.0119	0.0119	0.0119	0.0119
	TimesFM	0.0123	0.0123	0.0123	0.0122	0.0122	0.0121	0.0120	0.0121	0.0122	0.0121
	Naive	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186
	Sundial	0.0121	0.0120	0.0120	0.0119	0.0119	0.0117	0.0119	0.0119	0.0120	0.0118

Table 30: MAPE - Rq3 Covid Crash 2020 Log Pct

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	179.3309	153.5219	149.1637	152.7955	150.8495	156.1087	152.0301	137.9721	142.9641	139.2579
	Chronos	114.0797	110.8864	106.5589	104.8301	106.1903	105.9307	102.4651	103.7158	102.3686	103.5411
	TimesFM	136.8285	137.7110	140.3332	139.7699	139.3679	132.3071	128.8543	133.4775	138.1599	137.0197
	Naive	401.3719	401.3719	401.3719	401.3719	401.3719	401.3719	401.3719	401.3719	401.3719	401.3719
	Sundial	149.7961	153.2298	152.9125	128.7158	122.6833	139.2795	122.4161	139.3631	133.0985	127.9123

Table 31: MAE - Rq3 Covid Crash 2020 Log

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	1.8219	1.8182	1.8226	1.8147	1.8205	1.8187	1.8180	1.8078	1.8058	1.8039
	Chronos	1.9103	1.8889	1.8979	1.8996	1.9337	1.9749	2.0231	1.9791	1.9467	1.9175
	TimesFM	1.9194	1.9320	1.8823	1.8856	1.8613	1.8747	1.9189	1.8943	1.9110	1.8524
	Naive	1.8353	1.8353	1.8353	1.8353	1.8353	1.8353	1.8353	1.8353	1.8353	1.8353
	Sundial	1.8434	1.8606	1.8335	1.8205	1.8336	1.8117	1.8274	1.7918	1.8014	1.8215

Table 32: MAPE - Rq3 Covid Crash 2020 Log

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	1.1709	1.1685	1.1729	1.1664	1.1717	1.1709	1.1705	1.1641	1.1629	1.1616
	Chronos	1.2282	1.2149	1.2195	1.2212	1.2425	1.2696	1.3018	1.2732	1.2512	1.2320
	TimesFM	1.2358	1.2439	1.2099	1.2155	1.1984	1.2056	1.2349	1.2181	1.2297	1.1936
	Naive	1.1821	1.1821	1.1821	1.1821	1.1821	1.1821	1.1821	1.1821	1.1821	1.1821
	Sundial	1.1889	1.2001	1.1832	1.1749	1.1838	1.1681	1.1785	1.1536	1.1606	1.1730

Table 33: MAE - Rq3 Financial Crisis 2008 Linear Pct

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	0.0066	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058
	Chronos	0.0059	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058
	TimesFM	0.0060	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059
	Naive	0.0093	0.0093	0.0093	0.0093	0.0093	0.0093	0.0093	0.0093	0.0093	0.0093
	Sundial	0.0060	0.0059	0.0059	0.0060	0.0059	0.0059	0.0060	0.0060	0.0060	0.0059

Table 34: MAPE - Rq3 Financial Crisis 2008 Linear Pct

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	175552792762.1094	98.6963	98.6963	4168675.488792.9917	3865001863615.5288	3630313414841.7583	3900284838173.8877	3769165240645.1318	3792486411018.8340	3675240228802.4114
	Chronos	137159995225.7991	1515778125168.1501	195487802136.8596	1474487517445.1714	187898203987.9734	16011728075.6075	2152121131415.7551	207244840913.8021	1975956773546.6316	2032086151194.3625
	TimesFM	409553620595.1006	454209172967.2422	3710452805807.9238	4968089424565.9229	3924218955636.9429	2496166223491.8208	3461458252637.6768	3136282595563.4967	2730308437701.0815	5376540831950.1078
	Naive	40768154149769.6797	40768154149769.6797	40768154149769.6797	40768154149769.6797	40768154149769.6797	40768154149769.6797	40768154149769.6797	40768154149769.6797	40768154149769.6797	40768154149769.6797
	Sundial	4133962077598.4824	4026561613484.0635	179679677017.4783	4602133995961.0664	20773581772588.1167	2134389841738.4967	2174518578243.6035	3637286494906.2065	3996448632160.8169	415805550986.5230

Table 35: MAE - Rq3 Financial Crisis 2008 Linear

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	0.1759	0.1734	0.1734	0.1738	0.1737	0.1737	0.1738	0.1737	0.1737	0.1736
	Chronos	0.1943	0.2040	0.2216	0.2437	0.2431	0.2173	0.2061	0.2029	0.2090	0.2155
	TimesFM	0.2098	0.1981	0.2025	0.1896	0.1897	0.1840	0.1883	0.1968	0.1885	0.1924
	Naive	0.1734	0.1734	0.1734	0.1734	0.1734	0.1734	0.1734	0.1734	0.1734	0.1734
	Sundial	0.1966	0.1959	0.1986	0.1846	0.1833	0.1876	0.1819	0.1788	0.1774	0.1774

Table 36: MAPE - Rq3 Financial Crisis 2008 Linear

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	0.5933	0.5846	0.5846	0.5862	0.5861	0.5860	0.5863	0.5859	0.5859	0.5858
	Chronos	0.6570	0.6899	0.7532	0.8309	0.8282	0.7382	0.6982	0.6869	0.7084	0.7311
	TimesFM	0.7115	0.6677	0.6806	0.6398	0.6399	0.6200	0.6330	0.6631	0.6345	0.6488
	Naive	0.5846	0.5846	0.5846	0.5846	0.5846	0.5846	0.5846	0.5846	0.5846	0.5846
	Sundial	0.6623	0.6603	0.6682	0.6236	0.6191	0.6336	0.6148	0.6050	0.5998	0.5998

Table 37: MAE - Rq3 Financial Crisis 2008 Log Pct

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	0.0066	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058
	Chronos	0.0059	0.0059	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058	0.0058
	TimesFM	0.0060	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059
	Naive	0.0093	0.0093	0.0093	0.0093	0.0093	0.0093	0.0093	0.0093	0.0093	0.0093
	Sundial	0.0059	0.0059	0.0059	0.0060	0.0059	0.0060	0.0059	0.0060	0.0059	0.0059

Table 38: MAPE - Rq3 Financial Crisis 2008 Log Pct

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	175552798762.1894	98.6063	98.6063	98.6063	98.6063	98.6063	4121957286762.2324	3570537235711.2075	3812285791045.2661	370565289843.7485
	Chronos	117719995237.7991	1267791256309.5603	150602979666.6760	1559066499769.5273	1698628771511.7981	1115463720361.2760	1786213350499.9229	1551217268116.7725	1996115271012.8025	2294543874596.9570
	TimesFM	405053026568.1086	394117937776.1562	436952104825.1382	461791389571.8662	4686826849672.0171	5887511727395.1152	2662614867155.1943	2552632385739.0341	2867135434869.4932	2676696064747.8015
	Naive	407681541.69769.6797	407681541.69769.6797	407681541.69769.6797	407681541.69769.6797	407681541.69769.6797	407681541.69769.6797	407681541.69769.6797	407681541.69769.6797	407681541.69769.6797	407681541.69769.6797
	Sundial	4266524442275.6548	638967676842.7793	2760639591751.9697	5150808399765.0723	3547476941143.0596	2952748636547.6934	5610712240306.5645	2757406466073.4453	1485279709290.8777	2417416367920.5845

Table 39: MAE - Rq3 Financial Crisis 2008 Log

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	0.1759	0.1734	0.1734	0.1734	0.1734	0.1796	0.1738	0.1737	0.1737	0.1736
	Chronos	0.1943	0.2028	0.2002	0.2002	0.2162	0.2592	0.2487	0.2182	0.2022	0.2164
	TimesFM	0.2098	0.2006	0.2032	0.2005	0.1949	0.1862	0.1799	0.1883	0.1900	0.1895
	Naive	0.1734	0.1734	0.1734	0.1734	0.1734	0.1734	0.1734	0.1734	0.1734	0.1734
	Sundial	0.1976	0.1903	0.1972	0.1944	0.2024	0.1989	0.1815	0.1849	0.1805	0.1772

Table 40: MAPE - Rq3 Financial Crisis 2008 Log

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	0.5933	0.5846	0.5846	0.5846	0.5846	0.6067	0.5863	0.5860	0.5859	0.5858
	Chronos	0.6570	0.6865	0.6771	0.6769	0.7335	0.8848	0.8485	0.7413	0.6842	0.7342
	TimesFM	0.7115	0.6764	0.6865	0.6765	0.6568	0.6266	0.6059	0.6342	0.6393	0.6387
	Naive	0.5846	0.5846	0.5846	0.5846	0.5846	0.5846	0.5846	0.5846	0.5846	0.5846
	Sundial	0.6655	0.6420	0.6640	0.6552	0.6823	0.6718	0.6130	0.6246	0.6106	0.5990

Table 41: MAE - Rq3 Market Downturn 2022 Linear Pct

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059
	Chronos	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059
	TimesFM	0.0060	0.0060	0.0060	0.0059	0.0059	0.0058	0.0059	0.0059	0.0059	0.0059
	Naive	0.0085	0.0085	0.0085	0.0085	0.0085	0.0085	0.0085	0.0085	0.0085	0.0085
	Sundial	0.0061	0.0062	0.0062	0.0061	0.0059	0.0060	0.0060	0.0060	0.0060	0.0060

Table 42: MAPE - Rq3 Market Downturn 2022 Linear Pct

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	120.2759	112.7739	113.7098	115.6422	117.1553	115.3967	113.5850	112.4625	109.6816	111.0087
	TimesFM	137.8469	132.7509	127.2821	150.5924	119.5672	123.7718	124.5129	128.5831	130.8180	130.2251
	Naive	512.5199	512.5199	512.5199	512.5199	512.5199	512.5199	512.5199	512.5199	512.5199	512.5199
	Sundial	143.0596	152.5373	159.7478	164.1571	131.4554	142.6154	145.2898	123.8441	151.3344	154.9742

Table 43: MAE - Rq3 Market Downturn 2022 Linear

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353
	Chronos	1.9194	1.9326	1.9552	1.9908	1.9138	1.9106	1.9605	1.9638	1.9514	2.0185
	TimesFM	1.8230	1.8043	1.7974	1.8133	1.8942	1.9381	1.7588	1.7650	1.7480	1.7836
	Naive	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353
	Sundial	1.7878	1.7995	1.7893	1.7351	1.7205	1.7281	1.7493	1.7339	1.7290	1.7309

Table 44: MAPE - Rq3 Market Downturn 2022 Linear

Horizon	Method	25d	50d	75d	100d	125d	150d	175d	200d	225d	250d
1d	ARIMA	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897
	Chronos	0.6917	0.6972	0.7060	0.7197	0.6914	0.6907	0.7092	0.7111	0.7066	0.7324
	TimesFM	0.6588	0.6521	0.6491	0.6547	0.6844	0.7011	0.6350	0.6363	0.6309	0.6446
	Naive	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897
	Sundial	0.6466	0.6505	0.6464	0.6262	0.6208	0.6243	0.6314	0.6262	0.6246	0.6249

Table 45: MAE - Rq3 Market Downturn 2022 Log Pct

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059
	Chronos	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059	0.0059
	TimesFM	0.0060	0.0060	0.0060	0.0059	0.0059	0.0059	0.0059	0.0058	0.0059	0.0059
	Naive	0.0085	0.0085	0.0085	0.0085	0.0085	0.0085	0.0085	0.0085	0.0085	0.0085
	Sundial	0.0060	0.0061	0.0061	0.0059	0.0061	0.0060	0.0061	0.0060	0.0060	0.0061

Table 46: MAPE - Rq3 Market Downturn 2022 Log Pct

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000	100.0000
	Chronos	120.2759	117.5591	113.8721	112.9615	114.6701	117.7741	119.0336	115.7786	113.5888	111.0087
	TimesFM	137.8469	134.2514	158.1729	130.1143	123.7492	121.5397	127.1292	122.8455	128.0162	130.2251
	Naive	512.5199	512.5199	512.5199	512.5199	512.5199	512.5199	512.5199	512.5199	512.5199	512.5199
	Sundial	137.9820	148.3442	159.6735	139.5894	169.2580	133.6010	145.1057	133.0309	168.4936	148.2471

Table 47: MAE - Rq3 Market Downturn 2022 Log

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353
	Chronos	1.9194	1.9701	1.9447	1.9235	1.9673	1.9673	1.9402	1.9079	1.9673	2.0185
	TimesFM	1.8230	1.8421	1.8108	1.8187	1.7620	1.7972	1.8182	1.9403	1.7612	1.7836
	Naive	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353	1.6353
	Sundial	1.7854	1.7823	1.7793	1.7837	1.8101	1.7745	1.7316	1.7164	1.7320	1.7337

Table 48: MAPE - Rq3 Market Downturn 2022 Log

Horizon	Method	25d	32d	41d	53d	69d	89d	116d	149d	193d	249d
1d	ARIMA	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897
	Chronos	0.6917	0.7101	0.7019	0.6939	0.7106	0.7106	0.7011	0.6899	0.7123	0.7324
	TimesFM	0.6588	0.6657	0.6536	0.6564	0.6362	0.6480	0.6569	0.7005	0.6354	0.6446
	Naive	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897	0.5897
	Sundial	0.6455	0.6443	0.6429	0.6452	0.6549	0.6414	0.6247	0.6197	0.6254	0.6261